

Complex Analysis Homework 8

张溢致

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1. Suppose that a meromorphic function f has two periods ω_1 and ω_2 , with $\omega_2/\omega_1 \in \mathbb{R}$.

(a) Suppose ω_2/ω_1 is rational, say equal to p/q , where p and q are relatively prime integers. Prove that as a result the periodicity assumption is equivalent to the assumption that f is periodic with the simple period $\omega_0 = \frac{1}{q}\omega_1$.

[Hint: Since p and q are relatively prime, there exist integers m and n such that $mq + np = 1$ (Corollary 1.3, Chapter 8, Book I).]

(b) If ω_2/ω_1 is irrational, then f is constant. To prove this, use the fact that $\{m - n\tau\}$ is dense in $\mathbb{R}$ whenever τ is irrational and m, n range over the integers.

证明 (a) 因为 p, q 互质, 存在整数 m, n 使得

$$mq + np = 1.$$

由 $\omega_2 = \frac{p}{q}\omega_1$, 得 $q\omega_2 = p\omega_1$ 。于是

$$f(z + \omega_0) = f\left(z + \frac{1}{q}\omega_1\right) = f\left(z + (mq + np)\frac{1}{q}\omega_1\right) = f(z + m\omega_1 + n\omega_2) = f(z).$$

故 ω_0 是 f 的周期, 双周期性等价于单周期性。

(b) 不妨设 $\omega_1 = 1$, $\tau = \omega_2 \notin \mathbb{Q}$ 。则对任意整数 k, l , $k + l\tau$ 都是周期。已知无理数的整数组合 $\{k + l\tau\}$ 在 $\mathbb{R}$ 中稠密, 从而周期在实轴方向稠密。对任意 $z, w \in \mathbb{C}$, 可取周期序列 $t_n \rightarrow w - z$, 则

$$f(w) = f(z + t_n) \rightarrow f(z),$$

故 f 恒为常数。

2. Suppose that $a_1, \dots, a_r$ and $b_1, \dots, b_r$ are the zeros and poles, respectively, in the fundamental parallelogram of an elliptic function f . Show that

$$a_1 + \dots + a_r - b_1 - \dots - b_r = n\omega_1 + m\omega_2$$

for some integers n and m .

[Hint: If the boundary of the parallelogram contains no zeros or poles, simply integrate $zf'(z)/f(z)$ over that boundary, and observe that the integral of $f'(z)/f(z)$ over a side is an integer multiple of $2\pi i$. If there are zeros or poles on the side of the parallelogram, translate it by a small amount to reduce the problem to the first case.]

证明 若原基本平行四边形 ∂P 的边上有 f 的零点或极点, 取一个充分小的复数 δ , 使得平移后的平行四边形

$$P_\delta = P + \delta$$

的边界 ∂P_δ 上不再含有 f 的任何零点与极点, 同时 f 在 P_δ 内部的零点、极点与 P 内部的零点、极点完全相同。

设 f 为椭圆函数, 在 z_0 处为 n 阶零点, 则

$$\frac{f'(z)}{f(z)} = \frac{n}{z - z_0} + G(z),$$

其中 $G(z)$ 是在 z_0 处全纯的函数。因此

$$\frac{zf'(z)}{f(z)} = \frac{nz}{z - z_0} + zG(z),$$

从而 $\frac{zf'(z)}{f(z)}$ 在 z_0 处的留数为 nz_0 。

同理, 若 z_0 是 n 阶极点, 则留数为 $-nz_0$ 。因此有

$$\frac{1}{2\pi i} \int_C \frac{zf'(z)}{f(z)} dz = a_1 + \cdots + a_r - b_1 - \cdots - b_r,$$

其中 $C = C_1 \cup C_2 + \omega_2 \cup C_1 + \omega_1 \cup C_2$ 为某一基本平行四边形的边界。

另一方面, 考虑对边的两段积分:

$$\left(\int_{C_1} + \int_{C_1 + \omega_1} \right) \frac{zf'(z)}{f(z)} dz = - \int_{C_1} \omega_1 \frac{f'(z)}{f(z)} dz,$$

而

$$- \int_{C_1} \frac{f'(z)}{f(z)} dz = \log f(z_0 + \omega_2) - \log f(z_0),$$

其中 $f(z_0 + \omega_2) = f(z_0)$, 故

$$\log f(z_0 + \omega_2) - \log f(z_0) = 2n\pi i, \quad n \in \mathbb{Z}.$$

同理

$$\left(\int_{C_2} + \int_{C_2 + \omega_2} \right) \frac{zf'(z)}{f(z)} dz = 2m\pi i \omega_2, \quad m \in \mathbb{Z}.$$

结合前面所述, 有

$$\begin{aligned} a_1 + \cdots + a_r - b_1 - \cdots - b_r &= \frac{1}{2\pi i} \int_C \frac{zf'(z)}{f(z)} dz \\ &= \frac{1}{2\pi i} (2n\pi i \omega_1 + 2m\pi i \omega_2) \\ &= n\omega_1 + m\omega_2. \end{aligned}$$

3. In contrast with the result in Lemma 1.5, prove that the series

$$\sum_{n+m\tau \in \Lambda^*} \frac{1}{|n+m\tau|^2} \quad \text{where } \tau \in \mathbb{H}$$

does not converge. In fact, show that

$$\sum_{1 \leq n^2 + m^2 \leq R^2} 1/(n^2 + m^2) = 2\pi \log R + O(1) \quad \text{as } R \rightarrow \infty.$$

证明 当 R 足够大时, 将格点和与二重积分比较:

$$\iint_{1 \leq x^2 + y^2 \leq R^2} \frac{dxdy}{x^2 + y^2} \leq \sum_{1 \leq n^2 + m^2 \leq R^2} \frac{1}{n^2 + m^2} \leq \iint_{x^2 + y^2 \leq 2R^2} \frac{dxdy}{x^2 + y^2}.$$

极坐标下

$$\int_0^{2\pi} \int_1^R \frac{1}{r^2} \cdot r dr d\theta = 2\pi \int_1^R \frac{1}{r} dr = 2\pi \log R.$$

因此和为 $2\pi \log R + O(1)$, 当 $R \rightarrow \infty$ 时趋于无穷, 原级数发散。

7. Setting $\tau = 1/2$ in the expression

$$\sum_{m=-\infty}^{\infty} \frac{1}{(m+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)},$$

deduce that

$$\sum_{m \geq 1, m \text{ odd}} \frac{1}{m^2} = \frac{\pi^2}{8} \quad \text{and} \quad \sum_{m \geq 1} \frac{1}{m^2} = \frac{\pi^2}{6} = \zeta(2).$$

Similarly, using $\sum 1/(m+\tau)^4$ deduce that

$$\sum_{m \geq 1, m \text{ odd}} \frac{1}{m^4} = \frac{\pi^4}{96} \quad \text{and} \quad \sum_{m \geq 1} \frac{1}{m^4} = \frac{\pi^4}{90} = \zeta(4).$$

These results were already obtained using Fourier series in the exercises at the end of Chapters 2 and 3 in Book I.

证明 代入 $\tau = 1/2$, 得

$$\sum_{m=-\infty}^{\infty} \frac{1}{(2m+1)^2} = \frac{\pi^2}{4},$$

即

$$\sum_{m \geq 1, m \text{ odd}} \frac{1}{m^2} = \frac{\pi^2}{8}$$

考虑

$$\begin{aligned} \sum_{m=1}^{\infty} \frac{1}{m^2} &= \sum_{m \geq 1, m \text{ odd}} \frac{1}{m^2} + \sum_{m \geq 1, m \text{ even}} \frac{1}{m^2} \\ &= \sum_{m \geq 1, m \text{ odd}} \frac{1}{m^2} + \frac{1}{4} \sum_{m=1}^{\infty} \frac{1}{m^2} \end{aligned}$$

从而即有

$$\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{4}{3} \sum_{m \geq 1, m \text{ odd}} \frac{1}{m^2} = \frac{\pi^2}{6}$$

对

$$\sum_{m=-\infty}^{\infty} \frac{1}{(m+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)},$$

中 τ 求两次导数, 得到

$$\sum_{m=-\infty}^{\infty} \frac{1}{(m+\tau)^4} = \frac{\pi^4}{3} \cdot \frac{2 + \cos^2(\pi\tau)}{\sin^4(\pi\tau)}.$$

代入 $\tau = 1/2$ 得

$$\sum_{m=-\infty}^{\infty} \frac{1}{(m+1/2)^4} = \frac{2\pi^4}{3},$$

从而

$$\sum_{\substack{m \geq 1 \\ m \text{ 奇}}} \frac{1}{m^4} = \frac{\pi^4}{96}$$

类似前面可以得到

$$\sum_{m=1}^{\infty} \frac{1}{m^4} = \frac{\pi^4}{90} = \zeta(4).$$

1. Besides the approach in Section 1.2, there are several alternate ways of dealing with the sum $\sum 1/(z+\omega)^2$, where $\omega = n + m\tau$. For example, one may sum either (a) circularly, (b) first in n then in m , (c) or first in m then in n .

(a) Prove that if $z \notin \Lambda$, then

$$\lim_{R \rightarrow \infty} \sum_{n^2+m^2 \leq R^2} \frac{1}{(z+n+m\tau)^2} = S_1(z)$$

exists and $S_1(z) = \wp(z) + c_1$.

(b) Similarly,

$$\sum_m \left(\sum_n \frac{1}{(z+n+m\tau)^2} \right) = S_2(z)$$

exists and $S_2(z) = \wp(z) + c_2$, where $c_2 = F(\tau)$, and F is the forbidden Eisenstein series.

(c) Also

$$\sum_n \left(\sum_m \frac{1}{(z+n+m\tau)^2} \right) = S_3(z)$$

exists with $S_3(z) = \wp(z) + c_3$, and $c_3 = \tilde{F}(\tau)$, the reverse of F .

[Hint: To prove (a), it suffices to show that $\lim_{R \rightarrow \infty} \sum_{1 \leq n^2+m^2 \leq R^2} 1/(n+m\tau)^2 = c_1$ exists. This is proved by a comparison with $\int_{1 \leq x^2+y^2 \leq R^2} \frac{dx}{(x+y\tau)^2} = I(R)$. It can be shown that $I(R) = 0$, which follows because $(x+y\tau)^{-2} = -(\partial/\partial x)(x+y\tau)^{-1}$.]

证明 (a) 记

$$S_1(R, z) = \sum_{n^2+m^2 \leq R^2} \frac{1}{(z+\omega)^2}.$$

$\wp$ 可写为

$$\wp(z) = \frac{1}{z^2} + \lim_{R \rightarrow \infty} \sum_{\substack{(n,m) \neq (0,0) \\ n^2+m^2 \leq R}} \left(\frac{1}{(z+n+m\tau)^2} - \frac{1}{(n+m\tau)^2} \right).$$

则

$$S_1(R, z) = \wp(z) + \lim_{R \rightarrow \infty} \sum_{1 \leq n^2+m^2 \leq R^2} 1/(n+m\tau)^2$$

由于 $\sum_{1 \leq n^2+m^2 \leq R^2} 1/(n+m\tau)^2$ 不绝对收敛, 故需判断其按照圆形求和时是否收敛

记

$$S(R) = \sum_{1 \leq n^2 + m^2 \leq R^2} \frac{1}{(n + m\tau)^2}.$$

定义积分

$$I(R) = \iint_{1 \leq x^2 + y^2 \leq R^2} \frac{1}{(x + y\tau)^2} dx dy.$$

由提示, 注意到

$$\frac{1}{(x + y\tau)^2} = -\frac{\partial}{\partial x} \left(\frac{1}{x + y\tau} \right),$$

即被积函数是关于 x 的全微分。对环形闭区域应用格林公式, 其边界积分等于 0, 因此

$$I(R) = 0.$$

于是

$$S(R) = S(R) - I(R).$$

对每个格点 (n, m) , 将和式与单位正方形上的积分比较:

$$\left| \frac{1}{(n + m\tau)^2} - \int_n^{n+1} \int_m^{m+1} \frac{1}{(x + y\tau)^2} dx dy \right| \leq \frac{C}{|n + m\tau|^3},$$

其中 C 是仅依赖 τ 的常数。

对区域 $1 \leq n^2 + m^2 \leq R^2$ 求和得

$$|S(R) - I(R)| \leq C \sum_{1 \leq n^2 + m^2 \leq R^2} \frac{1}{|n + m\tau|^3}.$$

由于级数 $\sum \frac{1}{|n + m\tau|^3}$ 绝对收敛, 故当 $R \rightarrow \infty$ 时, 上式右边趋于一个有限极限。

因为 $I(R) = 0$, 所以 $S(R)$ 必收敛于有限极限, 即

$$c_1 = \lim_{R \rightarrow \infty} \sum_{1 \leq n^2 + m^2 \leq R^2} \frac{1}{(n + m\tau)^2}$$

存在, 且 c_1 仅依赖于 τ 。

于是 $c_1 = \lim_{R \rightarrow \infty} \sum_{1 \leq n^2 + m^2 \leq R^2} 1/(n + m\tau)^2$ 存在, 于是得证

(b)

定义和式:

$$S_2(z) = \sum_{m \in \mathbb{Z}} \left(\sum_{n \in \mathbb{Z}} \frac{1}{(z + n + m\tau)^2} \right).$$

将 $\wp(z)$ 的定义按先 n 后 m 的次序拆分:

$$\wp(z) = \frac{1}{z^2} + \sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} \left(\frac{1}{(z + n + m\tau)^2} - \frac{1}{(n + m\tau)^2} \right),$$

将 $S_2(z)$ 与 $\wp(z)$ 作差:

$$S_2(z) - \wp(z) = \sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} \frac{1}{(n + m\tau)^2}.$$

记该双重级数为 $F(\tau)$, 因此:

$$S_2(z) = \wp(z) + F(\tau).$$

(c)

定义和式:

$$S_3(z) = \sum_{n \in \mathbb{Z}} \left(\sum_{m \in \mathbb{Z}} \frac{1}{(z + n + m\tau)^2} \right).$$

将 $\wp(z)$ 的定义按先 m 后 n 的次序拆分:

$$\wp(z) = \frac{1}{z^2} + \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} \left(\frac{1}{(z + n + m\tau)^2} - \frac{1}{(n + m\tau)^2} \right).$$

将 $S_3(z)$ 与 $\wp(z)$ 作差:

$$S_3(z) - \wp(z) = \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} \frac{1}{(n + m\tau)^2}.$$

记该双重级数为 $\tilde{F}(\tau)$, 因此:

$$S_3(z) = \wp(z) + \tilde{F}(\tau).$$

2. Show that

$$\wp(z) = c + \pi^2 \sum_{m=-\infty}^{\infty} \frac{1}{\sin^2((z + m\tau)\pi)}$$

where c is an appropriate constant. In fact, by part (b) of the previous problem $c = -F(\tau)$.

证明 取上一题 (b), 有

$$S_2(z) = \sum_{m \in \mathbb{Z}} \left(\sum_{n \in \mathbb{Z}} \frac{1}{(z + n + m\tau)^2} \right) = \sum_{m \in \mathbb{Z}} \frac{\pi^2}{\sin^2((z + m\tau)\pi)} = \wp(z) + F(\tau)$$

从而

$$\wp(z) = c + \pi^2 \sum_{m=-\infty}^{\infty} \frac{1}{\sin^2((z + m\tau)\pi)}$$